Adrith Kurlagunda

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EDUCATION

University of California, Los Angeles (UCLA) — **Los Angeles, CA**
B.S. in Computer Engineering — *Expected June 2030*

Irvington High School — **Fremont, CA**
GPA: 4.00/4.00 | SAT: 1560 | National Merit Scholarship Finalist — *June 2026*

EXPERIENCE

COSMOS Summer Program, University of California, Santa Cruz — **Santa Cruz, CA**
Student Researcher — *August 2025 - September 2025*
- Compared random forest, logistic regression, and support vector machine learning models for electrical-grid failure prediction
- Assessed accuracy and cost-effectiveness across all 3 approaches and presented the findings to 15+ UC Santa Cruz professors

Xponent Power — **Fremont, CA**
Engineering Intern — *July 2025 - August 2025*
- Assembled solar-system subassemblies for deployable solar awnings, supporting both prototypes and manufacturing projects
- Verified 30+ components against CAD specifications and worked with engineers to integrate them into awning prototypes
- Documented recurring assembly issues and organized feedback from prototype builds to help clarify assembly instructions

D2Solar — **San Jose, CA**
Engineering Intern — *June 2024 - August 2024*
- Designed and fabricated an adjustable fixture for electroluminescence testing, reducing solar-panel alignment time by ∼25%
- Created 3 production soldering jigs and 2 CNC-ready drawings; supported manufacturing and assembly of 120+ solar cells
- Coordinated with CNC machining vendors to manufacture custom aluminum components for the solar-panel testing fixture

PROJECTS

Autonomous Indoor Mapping Robot — *GitHub*
C++, Python, ROS 2, ESP32, Raspberry Pi 5, SLAM, Nav2
- Built a differential-drive robot that autonomously explored and mapped 1,200+ sq ft of indoor space using LiDAR SLAM
- Programmed ESP32 firmware in C++ for motor control, encoder odometry, sensor data acquisition, and PID velocity control
- Used SLAM Toolbox and Nav2 with wheel odometry in ROS 2 to generate 2D maps and plan autonomous navigation paths

Fresnel-Lens Solar Water Purification System — *Patent*
U.S. Patent US12528717B1
- Designed a portable Fresnel-lens water purification system, refining CAD models and 3D-printed prototypes over 4 years
- Integrated Arduino temperature and pressure sensors with solenoid flow control; purified water in India within 6 minutes
- Authored the U.S. patent application through USPTO grant for a collapsible solar-thermal system with passive sun-tracking

Relist: AI-Powered Resale Listing Assistant — *GitHub*
Python, FastAPI, Next.js, React, TypeScript, OpenAI API, OpenCV
- Built Relist, an AI-powered app that converts clothing photos into editable resale listings and Depop-compatible CSV files
- Built a FastAPI backend using OpenCV for image-quality checks and the OpenAI API for garment analysis and pricing
- Developed a React and Next.js frontend for reviewing AI-generated listings, editing product details, and approving exports

LEADERSHIP

Varsity Science Olympiad — **Fremont, CA**
Director of Internal Affairs — *2022 - 2026*
- Organized practice sessions for a 20+ member varsity team and mentored students in robotics, circuits, chemistry, and CAD
- Coordinated with tournament organizers and other teams to confirm registration requirements and competition logistics

ExpertiseNPO, Educational Nonprofit — **Fremont, CA**
External Vice President — *2024-2026*
- Led outreach to 10+ industry professionals to recruit guest speakers for nonprofit educational events serving 500+ students
- Coordinated guest speakers and schedules across 6 educational events, supporting event planning and program delivery

ADDITIONAL INFORMATION

Skills: Python, SQL, C++, JavaScript, React, AI Workflow Automation, Git/GitHub, Fusion 360, SolidWorks, 3D printing
Interests: Spikeball, Specialty Coffee, Working Out, UFC, Vintage Clothing, Porsches, Beach Days, Tesla FSD, Underground Rap